o General Science

WSN

1

Important Topic & Questions

Unit - 1 & Unit - 2 Based on Aakash

Adhoc N/w
→ Characteristics
→ features
→ Architecture
→ MAC Protocol
→ Appⁿ

Cellular v/s ad-hoc N/w

Types of Routing protocols
EER
GR

Wireless Sensor Network
→ Architect. → Def → components → operations

Contention based protocols

Power Management in ad-hoc N/w

Transport layer protocol for AdHoc N/w

~~Single hop & Multi hop N/w with Diagram~~

Unit-1

- [x] MANETS. (Def., Charact., Challenges, properties, Applications)
- [x] Sensor N/w (Constraints, challenges, Advantages, Applications Architecture)
- [x] Energy Consumption of Sensor Nodes.
- [x] Operating Systems
- [x] N/w Architecture (Sensor N/w Scenarios)
- [x] Optimization Goals
- [x] Gateway Concepts

Unit-2

- [x] Networking Sensors (Physical layer & Transceiver design)
- [x] MAC Protocols for WSN (Classification, for Sensor N/w)
- [x] Location discovery
- [x] S-MAC
- [x] IEEE 802.15.4
- [ ] Routing Protocols (Energy Efficient Routing, Geographic Routing)

Wireless Sensor Network

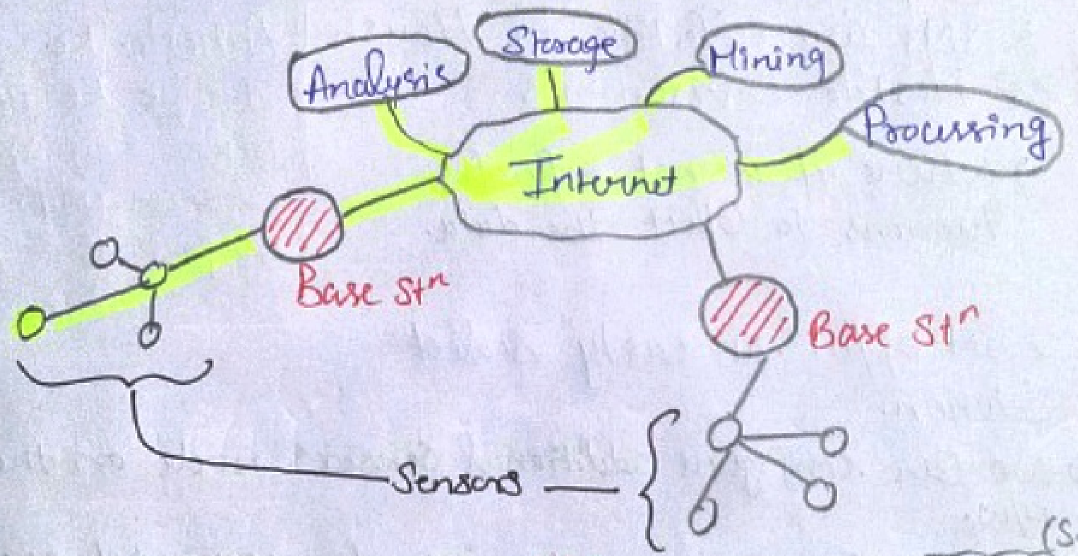
Phy Qty. → Sensor → Observable qty

Sensor N/W are highly distributed, lightweight nodes, deployed in large no. to monitor the environment or system.

→ Deployed in Ad-Hoc Manner — Latin :- "for this purpose"

↳ Sensor nodes are fitted with on board processor.

Sensor Node
→ Sensor Subsystem
→ Processing system
→ Comm^n System



★ Applications (BANPAI)

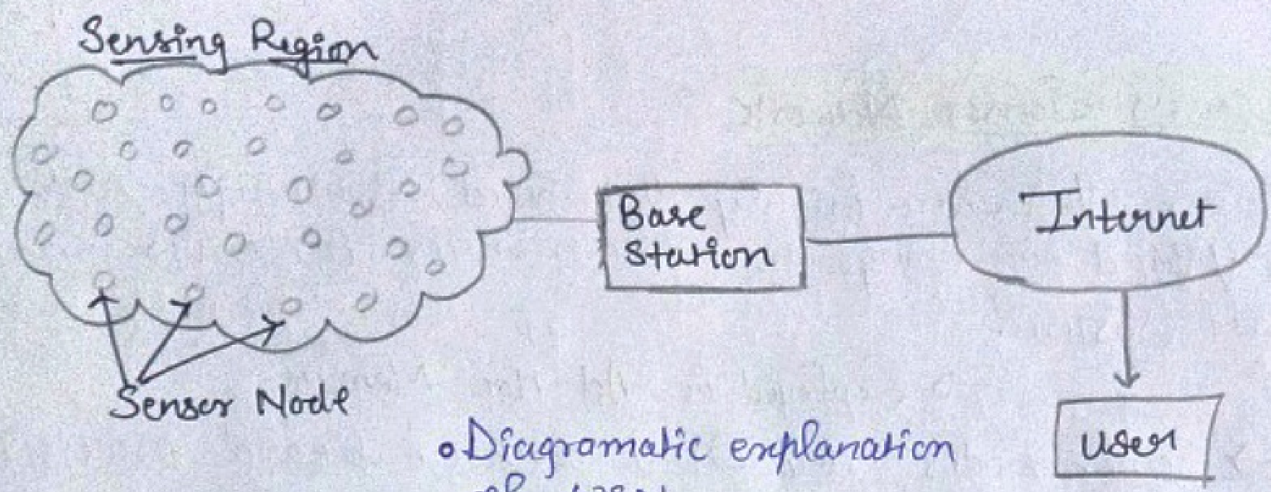
- (i) Battlefield - Surveillance & Monitoring
- (ii) Air pressure, Temp. Humidity
- (iii) Noise level
- (iv) Patient diagnosis & monitoring
- (v) Agriculture
- (vi) IOT
- (vii) Landslide Mon.
- (viii) water quality Mon.
- (ix) Health
- (x) Traffic

★ Challenges (SQES) (Squeeze)

- (i) Scalability → N/W ↑ Throughput ↓
- (ii) Quality of Service
- (iii) Energy efficiency
- (iv) Security

Proper Def.

Wireless Sensor N/W :- It refers to a group of spatially dispersed and dedicated sensors for monitoring and recording the physical conditions of the environment and organizing the collected data at a central location. WSN measure environmental conditions like temp., sound, pollution levels, humidity, wind speed & direction, pressure etc.



• Diagramatic explanation of WSN

Advantages :-

1. WSNs are effective in Harsh or Hostile Enviroments
   → Where it is not safe for humans to collect the data.

Pahado Ke upar
Jungle Ke paar
Dekho Kon apne
Shahar aaya yaar
~~Ninja Hattori~~ WSN

2. WSN offer an easily scaled Solution.
   → We can configure additional sensors node on the same N/W.
   → Can deploy sensor nodes into the expanded area.
   → No need to change or modify.

3. WSN enables long-distance data collection and transmission.

4. WSN can Anticipate Natural Disasters.
   → Sensors can predict or atleast can try to get the knowledge of changes in the environment.

5. WSNs can protect Hardware and Data Assets.

6. Landslide detection   7. Water quality Monitoring

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* Disadvantages of WSN

1. Easy to hack.
The simple functioning of sensor node devices makes these N/w easy to deploy, but it also makes them vulnerable to malicious security attacks, as these sensors often lack any robust security systems.

2. Comparatively low speed of communication.

3. Gets distracted by various elements like Bluetooth.

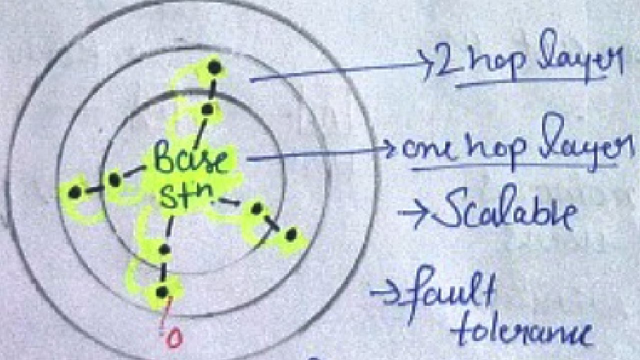
4. Costly.

5. Life of Nodes is a concern.

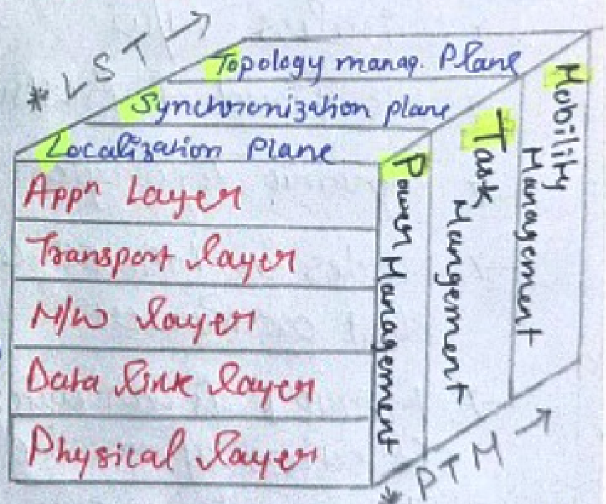
6. Energy life

Architecture of Sensor N/w

(i) Layered N/w Architecture :-
It includes five layers and three cross layers.

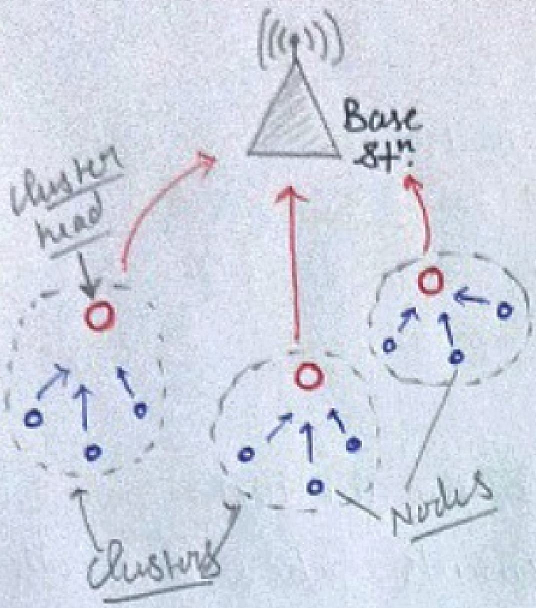


→ Scalable
→ fault tolerance
→ Power Consumption is less



→ Sensor N/w protocol Task

(ii) Clustered N/w Architecture :-



LEACH Protocol : Low Energy Adaptive Clustering Hierarchy

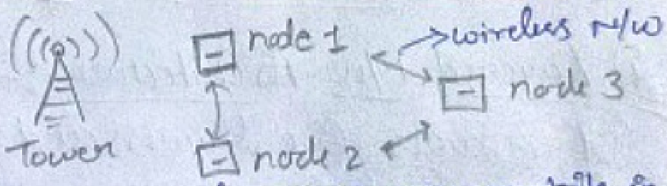
- 2-tier hierarchy clustering Arch.
- Distributed algorithm to organise sensor nodes into clusters.
- Cluster node heads create TDMA (time division multiple access) schedules
- Energy effitient = Data fusion

So, it has 2 phases :-

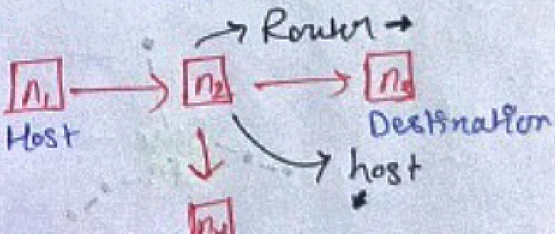
(1) Setup phase (2) Steady phase
→ Transmission of data

Mobile AdHoc Network (MANET)

- Wireless N/w
- No fixed infrastructure
- Dynamic Topologies
- Nodes in MANET can act as Host or Router
- MANET is an autonomous collection of mobile users that communicate over wireless links.



→ nodes are mobile in nature.



MANET →
- Mobile in nature
- No fixed Infrastructure
- Decentralized

★ Characteristics :-

1. Dynamic Topology : ever changing N/w structure within a MANET.

2. Energy Constrained nodes: No backup for energy as such. Nodes operate on battery power.

3. Limited Security : Sine nodes vastly have very limited source of energy, resources, memory and processing power, it is very difficult to build a robust security mechanism.

4. Autonomous : They don't have a central authority, they are self organized, have dynamic topology and do most of the work by themselves.

5. Distributed : They spread across a n/w and are even expandable.

★ Properties :- 1. Fast N/w establishment.

2. Peer-to-Peer Connectivity.

3. Independent Computation.

4. No requirement of access points.

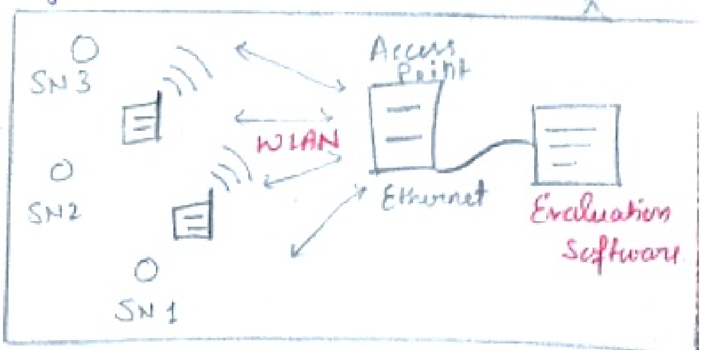
5. Less wireless connectivity range

★ **Challenges:-** 1. It has Dynamic Topology
2. Limited Security
3. Limited Bandwidth
4. Constrained energy
5. Routing is difficult (Because of mobility)

★ **Applications:-**
1. Battlefields : war zone (less time taken, easy infrastructure).
2. Sensor N/w (All Appⁿ of WSN)
3. VANET (Vehicular Ad Hoc N/w) TESLA
4. PAN (Personal Area N/w)

Components of WSN :-

(i) **Sensors** :- They capture the measured variable data, for further processing sensor signal is converted into electrical signals.



(ii) **Wireless Sensor node / Radio nodes** :- They receive the sensor data from sensors.

(iii) **WLAN Access Point** : Receives sensor data which are transferred by sensor nodes wirelessly.

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Components of wireless sensor "Nodes"

(i) **Micro Controllers**:- This is a Computer-on-a-chip which is very small in size.

→ Capable of doing powerful tasks including Controlling the funcn. of other devices Connecting to it.

Microcontroller →
- Microprocessor
- RAM Memory
- Associated Peripherals

(ii) **Transreceiver**:- This is a transmitter-receiver that is used for Commn purposes to send and receive data.

(iii) **External memory**:- WSN nodes usually use flash memories.
- a non-volatile memory that stores data and can be reprogrammed
- Small in size.
- Reasonable Storage Capacity.

(iv) **Power Source**:- Power is stored in the form of batteries.

Energy Consumption of Sensor Nodes :-

- Energy Consumption of a Sensor node must be tightly Controlled.
- The main Consumers of energy are the Controller, the radio front end, the memory and the types of Sensors.

To reduce power Consumption –

↳ 1st method :- Design low power chips.

↳ 2nd Method :- Reduce the functionality by using multiple states of operation

(i) Microcontroller energy Consumption :-

By using Normal Mode, idle Mode and Sleep Mode

	Normal Mode	idle Mode	Sleep Mode
	All work	Clock to the CPU stopped peripherals are active	RTC (Real time clock) remains active
Power Consumption →	$\boxed{400\,mW}$	$\boxed{100\,mW}$	$\boxed{50\,\mu W}$

(ii) Memory Energy Consumption :-

→ Energy Consumption is necessary for reading and writing of the flash memory.

→ writing of flash memory can be time Consuming and energy Consuming, it is best to avoid it if possible

(iii) Radio Transceivers energy Consumption :-

Transceivers –
- → Transmitting :- Turned off if not necessary
- → Receiving :- Turned off / ON.

(iv) Power Consumption of Sensor and actuators :-

Because of their wide variety, no guidelines as such.

Operating Systems :-

WSNs have limited processing power and memory, thus they require specialized OS and execution environments for their unique constraints.

Examples of OS for WSNs (Tiny OS Contiki RIOT Java Micro Python — TICORIJAMI)

- Tiny OS : → open source OS for WSNs.
  → nes C programming lang.
  → Lightweight & energy efficient.

- Contiki : → open source OS for WSNs.
  → C prog. lang.
  → Supports wide range of N/w protocols and Appⁿ
- RIOT : → open source OS for IoT
  → C prog. lang.
  → Lightweight & energy efficient.
- Java Embedded : → used to run Java appⁿ on WSN
  → Lightweight, energy efficient
  → Supports wide range of Java lib & frameworks.
- Micro Python :- → Python Prog. lang.
  → Lightweight
  → easy use & range of libraries.

Ti Co Ri Ja Mi

Acronym

★ Network Architecture & Sensor N/w Scenarios

→ Basic Scenarios in Wireless Sensor Networks

1. Types of Sources & Sinks.
2. Single-hop v/s Multihop N/w.
3. Multiple sinks & Sources.
4. Mobility ([Node, sink & Event] mobility)

1. Types of Sources & Sinks :-

→ **Sources** :- Any node that provides data / measurement or the actual nodes that sense data.

→ Nodes where info is generated.

→ **Sinks** :- Nodes where info is required.

→ Maybe inside / outside the Sensor N/w

→ it is a part of external N/w (e.g. Internet) Somehow Connected to the WSN

2. Single-hop v/s Multi-hop N/w :-

Single-hop

→ Simple & direct Commⁿ is not always possible in WSNs.

→ If area / ground is really big, it is difficult to Communicate.

→ they can't operate in difficult radio environment with strong attenuation.

→ Limited range of wireless Commⁿ.

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Multi-hop N/w

- node 1 → node 2 (intermediate node) → node (forwarding)
- Commⁿ in large distances or obstacles
- improved energy efficiency of Commⁿ
- proper store & forward should be there

3. Multiple Sinks, Multiple Sources :-

→ Multiple Sources should send info to multiple sinks
→ Either all or some of the info has to reach all or some of the sinks.

4. Mobility (Node, Sink & Event) :-

Since nodes are mobile, the WSN should have the ability to support Node, sink and event mobility.

Optimization Goals

1. Quality of Service
2. Energy efficiency
3. Scalability
4. Robustness

Quality of Service :-

- → Event detective
- → Event classification error
- → Event detection delay
- → Missing reports

Energy efficiency

- → Energy per correctly received
- → Energy for reported event
- → N/w life time
- → Time to loss of coverage

Scalability : Ability to maintain performance characteristics irrespective of the size of the N/w is referred to as scalability.

Robustness :- Wireless Sensor N/w should also exhibit an appropriate robustness.

→ They should not fail, just because a limited no. of nodes run out of energy, or something wrong happens b/w two nodes.

Gateway

In wireless sensor network, a gateway is a device that connects the sensor nodes to other networks, such as the internet.

→ It serves as an interface between the WSN and the outside world.

⇒ Key concepts related to gateway :-

- ☐ Commⁿ protocol : Gateway must be able to commⁿ with the sensor nodes using the appropriate commⁿ protocols.

- ☐ Data Aggregation : The gateway may be responsible for aggregating data from multiple sensor nodes before transmitting it to the outside world.

- ☐ Power Management : The gateway may need to manage the power consumption of the sensor node to ensure that they do not run out of power before their batteries can be replaced or recharged.

- ☐ Security : The gateway must be secured against attacks. This may involve encryption and other security measures.

- ☐ Deployment :- The gateway must be carefully deployed in the WSN to ensure that it has good coverage and can communicate effectively with the sensor nodes.

→ This may involve the use of multiple gateways to provide redundancy and improve stability.

Q) Why do we need a gateway ??

Ans-)
- ☐ Data transmission
- ☐ N/w management
- ☐ Security
- ☐ Interconnection
- ☐ Reliability.

Transceiver design Considerations in WSNs in a Physical layer

- Issues in Physical layer design in WSN
- Energy usage profile and the power Consumption in a transceiver.
- Choice of Modulation Scheme for the transceiver.
- Dynamic Modulation Scaling.
- Antenna Considerations.

→ Issues in Physical layer design in WSN

- There should be low power Consumption
- Due to Small/low power Consumption it is expected to have small transmit power and thus a small transmission range.
- As a further Consequence of low power Consumption, most hardware should be switched off or operated in a low-power standby mode most of the time.
- Comparatively low data rates per second required.
- Low implementation complexity and Cost.
- Low degree of Mobility.

→ Energy usage profile and the Power Consumption in a transceiver.

Transceivers Consume much more energy than they actually radiate.

→ Estimation is, a transceiver working at frequencies beyond 1 GHz takes 10 to 100 mW of power to radiate 1 mW energy.

- Many practical transmitter designs have efficiency below 10% at low radiated power.
- A second key observation is that for small transmit powers the transmit and receive modes consume more or less the same power.
- It is even possible that reception requires more power than transmission.
- idle mode → Energy Consumed
  thus Sleep state ✓ for low energy Consumpti
- A third key observation is the relative costs of Commⁿ v/s Computation in a sensor node
  → Clearly a comparison of these costs depends for the Commⁿ part on the bit rate error requirements, range, transceiver type etc.

⇒ <u>Choice of Modulation Scheme for the Transceiver</u>

- To maximize the time a transceiver can spend in sleep mode, the transmit times should be minimized.

→ The higher the data rate offered by a transceiver, the smaller the time needed to transmit a given amount of data and, consequently the smaller the energy consumption.

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→ Dynamic Modulation Scaling :-

→ It is interesting to consider methods to adapt the modulation scheme to the current situation. Such an approach is called dynamic modulation scaling.

⇒ Antenna Considerations :-

- The desired small form of nodes restricts the size and no. of antennas.
- If Antenna is smaller than carrier's wavelength, it is hard to achieve good antenna efficiency.
- The antennas should be spaced apart at least 40-50% of the wavelength used to achieve good effects from diversity.

★ MAC Protocols for Ad Hoc Wireless N/W

↳ Medium Access Control

↳ Responsible for transmitting data packets to and from one transmitting device to another device across shared channel.

Responsibilities of MAC :-

(i) Distributed MAC operation.

(ii) N/w overhead should be low.

(iii) Total BW must be allocated efficiently.

(iv) Maximize the utilization of channel.

(v) Good power control mechanism.

★★★ (vi) Hidden/exposed terminal probability should be eliminated.

(vii) Time Synchronization among nodes.

★ Design Issues of MAC Protocols → BW should be efficiently used

(i) Bandwidth Efficiency: Limited BW as radio spectrum is limited.

↳ $\boxed{\dfrac{\text{B/w used for actual data transmission}}{\text{Available BW}}}$ → Ratio

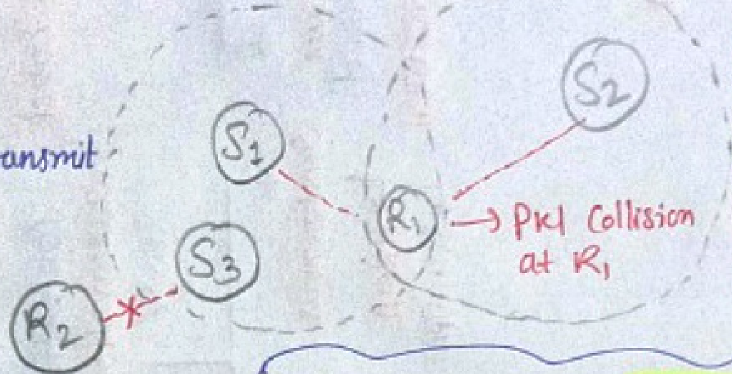
(ii) Quality of Service because of the mobility of nodes.

(iii) Synchronization :- Transmission b/w sender and receiver nodes has to be synchronized properly.

(iv) Hidden And Exposed Terminal Problem :-

* It is a major design issue.

S3 Can't transmit pkts to R2 because it thinks that another transmission is going b/w ($S_1 \rightarrow R_1$)



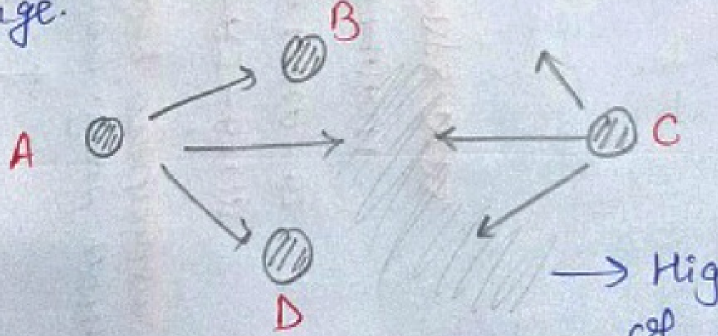
S_1 & S_2 are hidden from each other, Causing Collision

S_3 & S_1 are Exposed to each other

(v) Error Prone Shared Broadcast Channel :-

(Broadcast Nature of radio channel)

↳ Transmission made by a node are received by all nodes within its direct transmission range.



→ High possibility of pkt Collision.

(vi) Lack of Central Coordination : In Ad Hoc N/w nodes keep moving Continuously (distributed nature).

Additional overhead in terms of BW Consumption should be low. (Nodes Mobility should be taken care of

Classification of MAC Protocols (Imp)

MAC Protocol

- Contention based (QoS ✗)
  - Sender initiated
    - Single Channel
      - MACAW
    - Multi Channel
      - BTMA
  - Receiver initiated
    - RI BTMA
- Contention based with reservation mechanism (QoS ✓)
  - Synchronous
    - HRMA
    - SHMA/PA
  - Asynchronous
    - MACA/PR
    - RT MAC
- Contention based with Scheduling mechanism
  - RI-BTMA
  - MACA-BI
  - MARCH
  - DPS, DLPS
- Other MAC Protocols
  - MMAC
  - PCM
  - Directional Antennas

→ 2 nodes together try to transmit

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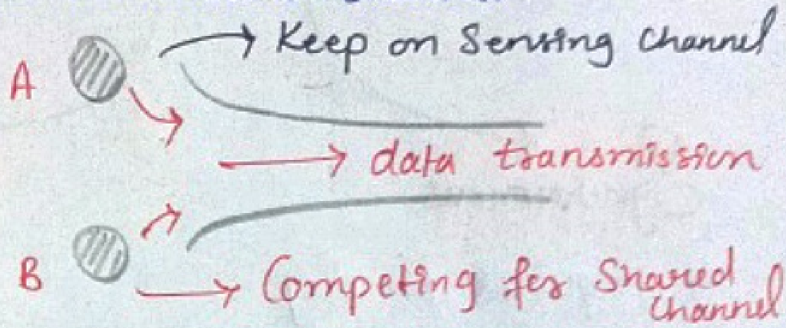
Contention - Based Protocols :- follows a Contention-based Channel access policy.

↳ No nodes make prior resource reservation.

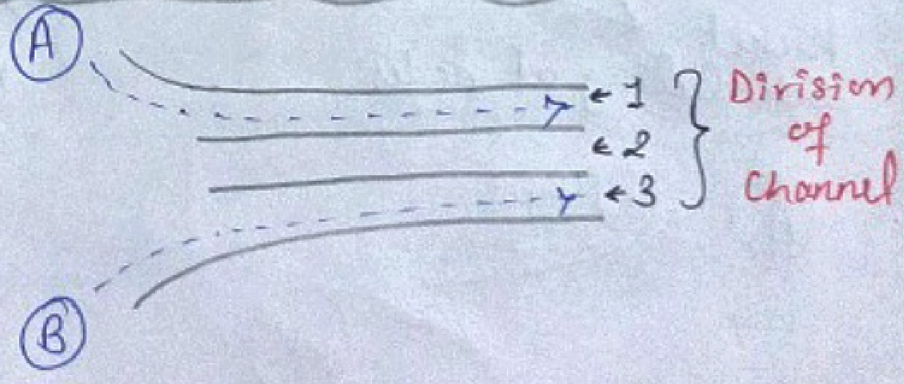
↳ When pkt is received, node competes with other neighbouring nodes to access channel.



⇒ Single - Channel Sender initiated :- Total available BW will not be divided into several channels.
[Node that wins channel uses entire BW].

⇒ Multi - channel Sender Initiated :- Total available BW is divided into multi-channels.
↳ Several nodes can simultaneously transmit data, each using a separate channel.

Receiver initiated — Receiver node initiates contention resolution protocols.



Contention-based Protocol with Reservation Mechanism :-

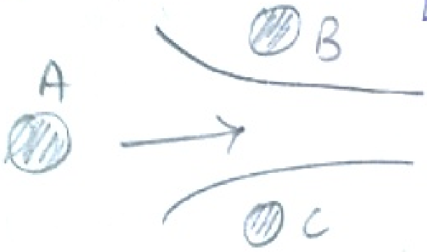
- Provides Support for real-time traffic.
- Provides QoS guarantee.
- Reserve BW prior to transmission.

Synchronous

It requires timely Synchronization among all the nodes.

Asynchronous

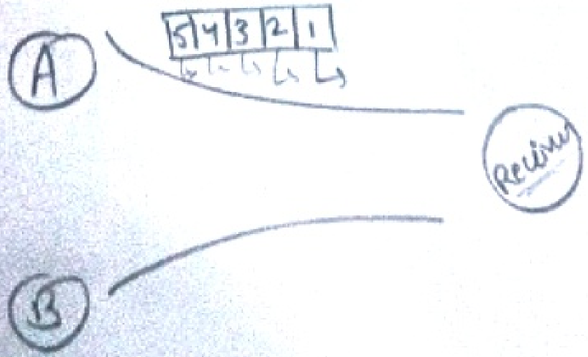
No requirement of Synchronization among nodes.
[relative time info^n]



Contention-based Protocols with Scheduling Mechanism.

Two operations

- Packet scheduling at nodes
- Scheduling nodes for channel access.



1st A then B or vice versa

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MAC Protocols for Sensor N/w :-

- Establish infrastructure for communication among sensor nodes.
- It shall manage power efficiency, topology change, global synchronization.

Types :- 1. Fixed - Allocation 2. Demand-Based 3. Contention - Based

(i) Fixed - Allocation :-

- Common medium is shared
- Appropriate for sensor N/w that continuously monitor & generate data traffic.
- Provided bounded delay for each node.

(ii) Demand - Based :-

- Channel is allocated according to node demand.
- Additional overhead of reservation process.
- Best suited for variable - rate traffic.

(iii) Contention - Based :-

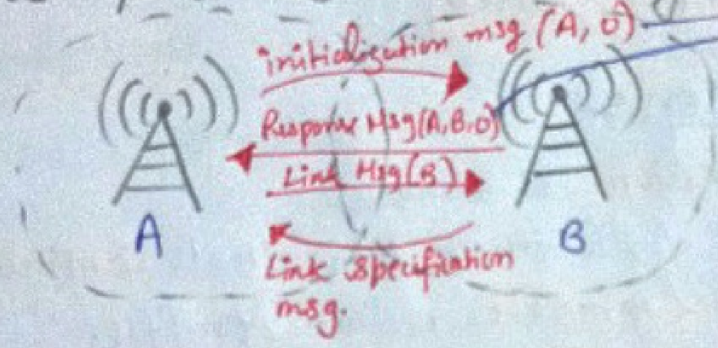
- Random access based contention for channel.
- Suitable for bursty traffic
- Possibility of collisions
- Not suitable for real-time traffic.

(i) Self-organizing MAC for Sensor N/W (SMACS) and Eavesdrop & Register (EAR)

N/W initialization — Mobility Support

**SMACS : Distributed Protocol

↳ Neighbour discovery and channel assignment takes place simultaneously.



→ No. of neighbours 'A' & 'B' initially 0

EAR :- Seamless Connection of Nodes under mobile & stationary Condition.

(ii) Hybrid TDMA/FDMA :- Centrally Controlled Scheme which assumes that nodes Communicate directly to a nearby Base St^n.

→ uses optimum no. of channels, which gives minimum overall power Consumption.

$$\Rightarrow \text{Ratio} = \frac{\text{Power Consumption by transmitter}}{\text{Power Consumption by receiver}}$$

$Tr > Re \rightarrow$ TDMA

$Re > Tr \rightarrow$ FDMA

(iii) CSMA-Based MAC Protocols :- In this Sensing periods of CSMA (carrier sense multiple access) are constant for energy efficiency.

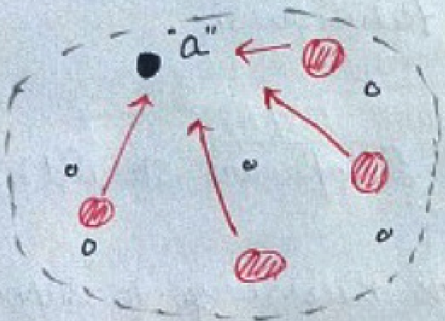
Backoff time is randomly decided to avoid collision.

Location Discovery :- Each node should know its location and manage to merge the information with the data in the message.

(i) **Indoor localization** :- uses fixed infrastructure to estimate the location of sensor nodes.

↳ fixed baecon nodes are placed strategically. ↳ they send the signals periodically

(literally → (guiding or warning signal) Baecon)

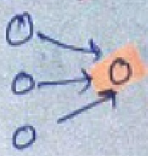


Signal Strength
↳ Angle of Arrival
↳ time difference b/w the signals arriving from different baecons.

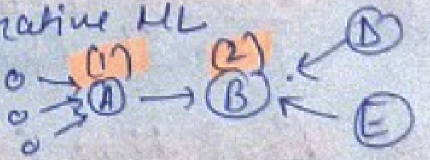
(ii) **Sensor N/w localization** :- No fixed infrastructure. Sensor nodes = Beacons | RF & Received Signal Strength indicator are used.

ML (Multi-lateration) :-

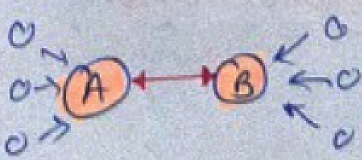
(i) Atomic ML:



(ii) Iterative ML



(iii) Collaborative ML



Evolving Standards :-

IEEE 802.15.4 → (LR-WPAN) → Low Rate wireless personal Area Network

- → operation under unlicensed frequency band.
- → Low data rate solution.
- → Multi year battery life with very low Complexity.

Applications → Home Automation
→ Remote Control

→ Its standard defines Physical / MAC layer specifications.

→ Low power consumption, transmission rate, power efficient modulation techniques.

Other Issues :-

(i) Energy efficient design :- Hardware (Microcontrollers) & Software (OS/Appⁿ) should be made energy efficient.

(ii) Synchronization :- Among nodes is necessary to support TDMA scheme on multihop n/w.
↳ Low power consumption in synchronization.

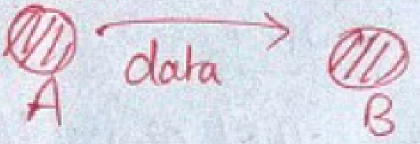
(iii) Transport layer issues :- (PSFQ) Pump slowly fetch quickly.
↳ Reliable data delivery, robust, scalable.

(iv) Security (LEAP) :- Localized encryption and authentication protocol (Key management).

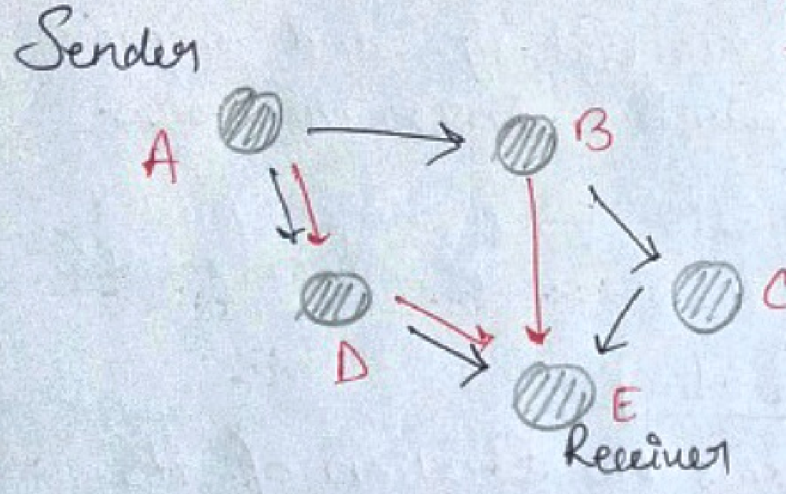
(v) Real time Communication

★ **Routing Protocols :-**

Routing is the process of establishing a path b/w the sender and receiver nodes for transmitting the packet along the path.



Routing protocol will find best optimum path
↳ Shortest path.



★ **Design Constraints :-**

(i) Mobility of nodes
(ii) Dynamic Topology
(iii) No centralized infrastructure
(iv) Bandwidth it is very less.
(v) Energy
(vi) Establishing end-to-end constraints.

★ **Characteristics :-** (i) fully distributed
(ii) Adaptive towards frequent changes.
↳ Mobility
(iii) Route computation and maintenance must involve minimum no. of nodes.
(iv) Packet collision must be minimum.

★ **Design issues :-**
(i) Mobility (ii) BW↓ (iii) Error prone shared channel
(iv) Hidden & exposed terminal (v) Resource ↓ (vi) Security↓

Energy Efficient Routing

↳ It is an algorithm that aims to reduce power usage, data latency and improve scalability in wireless sensor networks.

→ Gateway → Base st^n → manager nodes & sensor nodes

Geographical Routing

Each node knows the location of its direct neighbours (within Radio Range).
The source inserts the destination location inside the packet.

Great Great

Karan